

Catherine Fontaine

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EDUCATION

McGill University Master of Science (M.Sc.) in Mathematics and Statistics (Thesis)	09/2025 – 07/2027 GPA : 4.0
McGill University B.Sc. Joint Honours Mathematics and Computer Science Earned a First Class Honours Degree with Distinction	09/2021 – 04/2025 GPA : 3.91

PUBLICATION

L. Addario-Berry, C. Fontaine, R. Khanfir, L.-R.. Langevin, S. Têtu, (2024) *Optimal root recovery for uniform attachment trees and d-regular growing trees*, Accepted in Annals of Applied Probability.
<https://arxiv.org/abs/2411.18614>

RESEARCH PROJECTS

McGill University Department of Mathematics and Statistics <i>Master's Student</i> Project : Root-finding algorithms in trees generated by superlinear preferential attachment Supervisor : Prof. Louigi Addario-Berry • Impossibility result for root-finding algorithms in the superlinear preferential attachment model. • Study of the distance between the persistent hub and the root, leading to the development of a new approach better suited to the root-finding problem. • Preprint available soon on arXiv.	09/2025 –
McGill University Department of Mathematics and Statistics <i>Summer Undergraduate Researcher</i> Project : Root-finding algorithms in trees generated by superlinear preferential attachment Supervisor : Prof. Louigi Addario-Berry • Conducted a literature review on the superlinear preferential attachment model • Analyzed and synthesized existing results to identify open research directions	06/2025 – 08/2025
McGill University School of Computer Science <i>Research Project Course</i> Project : Clique Dimension in the Contradiction Graph of a Concept Class. Supervisor : Prof. Hamed Hatami • Investigate the clique dimension of contradiction graphs for various concept classes. • Attempt to improve the known upper bound as a function of the Littlestone dimension. • Tackle a problem introduced in “A Unified Characterization of Private Learnability via Graph Theory” by Alon, Moran, Scheffler, and Yehudayoff.	2024/09 – 2024/12
McGill University Department of Mathematics and Statistics <i>Summer Undergraduate Researcher</i> Project : Optimal root recovery for uniform attachment trees and d-regular growing trees. Supervisor : Prof. Louigi Addario-Berry	2024/05 – 2024/11

- Conduct research on root-finding algorithms in random growing trees generated using the uniform attachment model.
- Improve bounds from “Finding Adam in Random Growing Trees” by Bubeck, Devroye and Lugosi.
- Co-author a paper that outlines these findings, available on arXiv.

McGill University Department of Mathematics and Statistics

2023/05 – 2023/08

Summer Undergraduate Researcher

Project : Computing Counterexamples to Serre’s Modularity Conjecture

Supervisor : Prof. Patrick Allen

- Designed and implemented a Python algorithm using SageMath to compute counterexamples to a part of Serre’s Modularity Conjecture in the fields of modular forms and Galois representations.

SCIENTIFIC AND ACADEMIC TALKS

C. Fontaine (2026), *Where does a tree begin ? The Superlinear Case*,

Pizza Talk Mathematics and Statistics Graduate Students’ Association Seminar

C. Fontaine and al. (2026) *Orientation Presentation for New Undergraduate Students*,

McGill University.

C. Fontaine and al. (2026) *Presentation on Undergraduate Summer Research Award*,

McGill University.

C. Fontaine (2025), *Arbres aléatoires*,

Collège André-Grasset.

- Presentation to students in the Arts and Sciences program at Collège André-Grasset on mathematical research and an accessible overview of my research project as an example of contemporary mathematical research.

C. Fontaine, and S. Têtu (2025), *Identification de la racine d’un arbre généré par attachement uniforme*,

Les séminaires universitaires en mathématiques à Montréal (SUMM).

C. Fontaine and al. (2024) *Orientation Presentation for New Undergraduate Students*,

McGill University.

C. Fontaine (2024), *Where does a random tree begins ?*

McGill University.

C. Fontaine, R. Aron, Z. Horton (2023), *Computing Counterexamples to Serre’s Modularity Conjecture*,

McGill University

ACADEMIC AWARDS AND RESEARCH FUNDING

Bourse de maîtrise en recherche — FRQ 20,000 CAD, 2026

Lorne Trottier Science Accelerator Fellowships — McGill 5,000 CAD, 2025

Canada Graduate Scholarships — Master’s program — NSERC 27,000 CAD, 2025

Faculty of Science registration bonus — McGill 5,000 CAD, 2025

Grad Excellence Awards — McGill 2,870 CAD, 2025

Undergraduate Student Research Award — NSERC 8,950 CAD, 2025

Supplement to the Undergraduate Student Research Award — FRQ 1,500 CAD, 2025

Undergraduate Student Research Award — NSERC 8,700 CAD, 2024

Supplement to the Undergraduate Student Research Award — FRQ 1,500 CAD, 2024

Science Undergraduate Research Awards — McGill 8,350 CAD, 2023

TEACHING

Teaching Assistant for Math 141

01/2026 – 04/2026

Teaching Assistant for Math 140

09/2026 – 12/2026

COMPLEMENTARY TRAINING

ISM Discovery School

Université du Québec à Montréal, 06/2026

Recent advances in probability and combinatorics

- Attended conferences on coarse graph theory, the Kahn–Kalai conjectures, Ramsey theory, random Cayley graphs, and the cover time of random trees.

CRM-PIMS Summer School in Probability

University of British Columbia, 06/2025

- Attended conferences on percolation, heat kernel estimates and other topics in stochastic processes.

Directed Reading Program

McGill University, Supervisor : Tasmin Chu, 01/2025-06/2025

- A mentorship initiative pairing graduate and undergraduate students to collaborate on a supervised reading project.
- Studied Markov chains, random walks on graphs, and electrical network analogies (Levin & Peres, Lyons & Peres)
- Analyzed percolation and some branching processes

CRM-PIMS Summer School in Probability

Université de Montréal, 07/2024

- Attended conferences on random matrices, random walks and other topics in stochastic processes.

CRM-ISM Probability Seminar

McGill University, 09/2024-12/2024

Accelerated Introduction to ML

McGill Artificial Intelligence Society, 01/2023-04/2023

- One of twenty students selected to participate in a ten-week intensive course on machine learning.
- Implemented various machine learning algorithms including KNN, linear regression, naive Bayes, SVM and random forest.

SERVICE

Coordinator McGill Directed Reading Program in Mathematics

07/2025 –

- Organized a mentorship program pairing undergraduate and graduate students to collaborate on a supervised reading project.
- Coordinated communication and collaboration between participating undergraduate and graduate students.

VP Finance

07/2026 –

Graduate Student Association for Mathematics and Statistics at McGill University

- Preparation of the graduate students' association budget to organize events that foster connections among students and provide opportunities to present their research.

Mentor McGill Directed Reading Program in Mathematics

01/2026 – 04/2026

- Mentored two undergraduate students on the theory of Bienaymé trees (branching processes).

First-Year Student Representative

07/2025 – 04/2026

Graduate Student Association for Mathematics and Statistics, McGill **VP Events** — McGill Artificial Intelligence Society

11/2024 – 04/2025

- Organized MAIS Hacks, a 100-participant AI/ML hackathon featuring industry partnerships and technical workshops.

Podcast Producer — McGill Artificial Intelligence Society

04/2024 – 11/2024

- Planned podcasts featuring AI experts from academia and industry, exploring emerging trends and challenges in AI.

VP Event — McGill Artificial Intelligence Society

04/2023 – 04/2024

- Organized MAIS Hacks, a 150-participant AI/ML hackathon featuring industry partnerships and technical workshops.
- Organized MAIS Learnathon, a conference series with 80 participants, showcasing AI/ML experts.